CSE427: Machine Learning

Resource Sheet, Summer 2026

Mark Distribution:

Attendance - 10%

Quiz - 15% (Total 5 Class Tests and best 3 will be counted)

Midterm - 25%

Final - 30%

Lab - 20%

Consultation Timing: NWT_Consultation_Summer 2026.pdf *[Email before coming]*

Slides (Drive Link) : Summer 2026

Attendance & Marksheet:

Section 7: CSE427-07 Attendance & Marksheet M26

Section 8: CSE427-08 Attendance & Marksheet M26

Midterm Resources

Tentative Syllabus:

1. Introduction
2. Classification tree
3. Regression tree
4. Random Forest
5. AdaBoost
6. Logistic Regression
7. Gradient Boosting
8. Performance matrix of ML models, ROC curve
9. Dimension Reduction using PCA

Practice Problems: **CSE 427 practice problems for mid exam.pdf**

CSE427 Practice Problems for final.pdf[Performance matrix of ML models, ROC curve]

(These were given in Spring 2026)

Class	Topics	Slides	Hand Notes (may have errors) Or Recordings
1	Introduction	Lec_1a_ML.pptx	
2	Classification Trees	Lec_1b_Classification_tree.pptx	1.1 Classification Tree.pdf

3	Classification Tree & Regression Tree	Lec_2b_DT_Construction_using_inf… Lec_3a_RT.pdf lec_3b_step by step procedure for R…	1.2 Decision Tree.pdf CSE427: Decision Tree (ID3 Alg… [Classification Tree - ID3] 2.1 Regression Tree.pdf CSE427: Regression Tree (Spring…
4	Random Forest	Lec_4a_Random Forest.pdf Lec_4b_Random Forest.pptx	3.1 Random Forest.pdf 3.2 Finding Missing Value with Rand… CSE427: Random Forest and Fin…
5	AdaBoost	Lec_5_AdaBoost.pdf	4. AdaBoost.pdf
6	Logistic Regression Part I	Lec_9_Logistic Regression part 1.pptx Lec_10_logistic Regression Exampl…	7.1 Logistic Regression Part 1.pdf CSE427: Logistic Regression Exa… CSE427: Logistic Regression Exa…
7	Logistic Regression Part II	lec_7b_Logistic Regression Part II.p…	7.2 Logistic Regression Part II.pdf CSE427: Logistic Regression Part…
8	Gradient Boosting	lec_9_10_gradient boosting_regressi…	CSE427: Gradient Boosting Regr… CSE427: Gradient Boosting Classific…
9	Performance Evaluation & ROC-AUC	Lec_8_performance matrices_ROC.pdf	CSE427: Performance Evaluation…
10	PCA	PCA.pdf Lec_11_PCA_example_solution__st…	CSE427: PCA Concepts Math: https://youtu.be/VY1KB6HZeIs

Final Resources

Tentative Syllabus:

10. Neural network
11. Parameter updating of Neural network (Relu/Sigmoid activation function)
12. Parameter updating of neural network (Softmax function)
13. CNN
14. RNN
15. Word embedding
16. Transformer

Practice Problems: **CSE427 Practice Problems for final.pdf**

(It was given in Spring 2026)

CNN: **CNN_Math_Questions.pdf**

Class	Topics	Slides	Hand Notes (may have errors) Or Recordings
1	Neural network	lec_12_Neural network.pptx	Intro to Neural Network Recording by CMR sir Fall 2025
2	Parameter update in a neural network with Relu function at the output	Lec_13_parameter_update_with_RELU_function_at…	(Basics) Parameter U… Parameter Update Recording
3	Parameter update in a neural network with Softmax function at the output	lec_14_parameter_update2_softmax_function (1).pdf	No Notes or Recording :(
4	CNN	Lec_15_Convolutional Neural Networks.pdf	CNN Part 1 Recording CNN_without Norm…
5	RNN	Lec_17_18_Recurrent Neural Network.pdf	RNN & LSTM Recording
6	Word Embedding	Lec_21a_word_embedding.pdf	Watch StatQuest
7	Transformer	Lec_22a_positional_encoding.pdf Lec_22b_new_Transformer.pdf.pdf	Transformer Full Notes on Math: Transfomer Encoder.… Transfomer Decoder … Tasnfomer Deconder …

			Transformer - 1/2 Recording by CMR Sir Transformer - 2/2 Recording by CMR Sir